

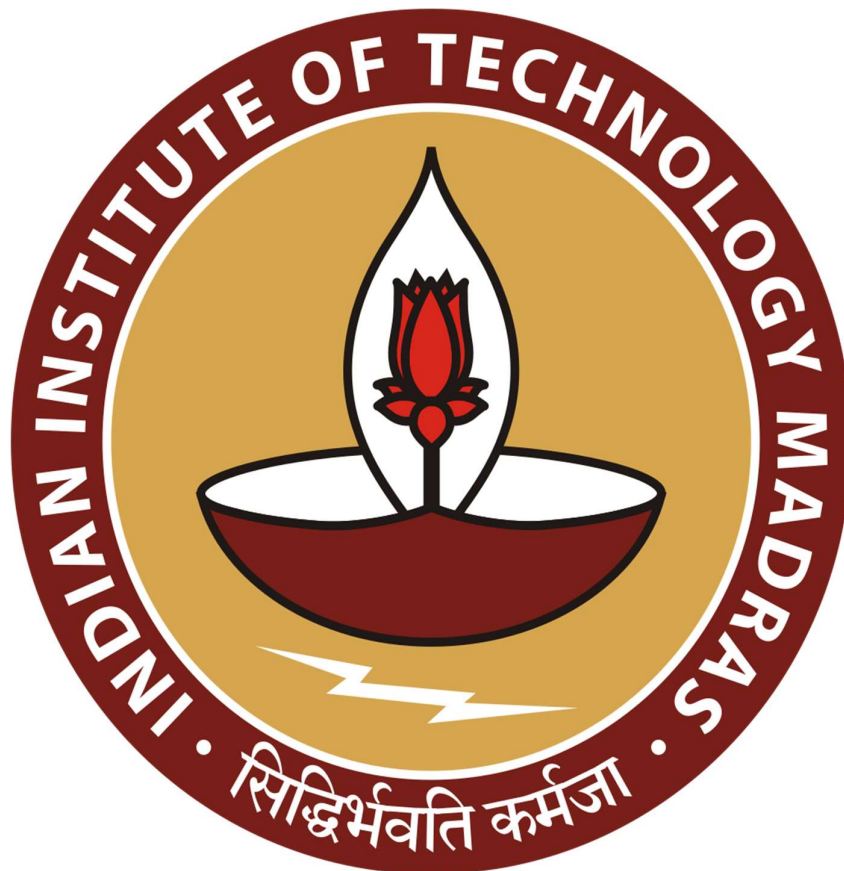
**Data-driven analysis and improvement strategies for Swiggy's
Customer satisfaction**

A Proposal report for the BDM capstone Project

Submitted by

Name: Ayush Vats

Roll number: 22f1000784



IITM Online BS Degree Program,
Indian Institute of Technology, Madras, Chennai
Tamil Nadu, India, 600036

Contents

- 1 Executive Summary and Title
- 2 Organisation Background
- 3 Problem Statement
 - 3.1 Problem statement 1
 - 3.2 Problem statement 2
 - 3.3 Problem statement 3
- 4 Background of the Problem (200 Words)
- 5 Problem Solving Approach (400 Words)
- 6 Expected Timeline
- 7 Expected Outcome

Declaration Statement

I declare that the project report titled "Data-Driven Analysis and Improvement Strategies for Swiggy's Customer Satisfaction" is my original work, conducted as part of my academic requirements at IIT Madras. The data utilised in this project has been sourced from Kaggle and was obtained through web scraping by the dataset creator.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. If plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfilment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Signature of Candidate: (Digital Signature)

Name - Ayush Vats

Date – 10 – 07 – 2024

Source of Data and Metadata:

The dataset used in this analysis was sourced from Kaggle, a platform for data science and machine learning. It was created via web scraping and comprises restaurant-related information from Swiggy. This dataset offers insights into performance metrics of diverse restaurants, essential for comprehending customer preferences and operational challenges in the food delivery sector.

Data Source link - <https://www.kaggle.com/datasets/abhijitdahatonde/swiggy-restuarant-dataset>

1. Executive Summary and Title:

Swiggy, founded in 2014, is India's leading food delivery platform, connecting millions of customers with a diverse range of restaurants nationwide. Swiggy primarily operates as a B2C (business-to-consumer) platform. It connects individual consumers with restaurants for food delivery services. The company is committed to providing a seamless and enjoyable food delivery experience, setting high standards in the industry.

This report addresses challenges faced by Swiggy in maintaining consistent customer satisfaction across different regions. Key issues include variations in service quality, delivery times, and pricing strategies, which impact overall customer experience and loyalty.

My approach involves analyzing Swiggy's restaurant dataset to identify and mitigate key operational challenges. Solutions include optimizing delivery logistics, refining pricing strategies based on customer preferences, and implementing targeted improvements in service quality across regions. By leveraging data-driven insights, Swiggy aims to enhance operational efficiency, improve customer satisfaction, and sustain its competitive edge in the food delivery market.

2. Organization Background:

Founded in 2014, Swiggy has rapidly evolved into a leading force in India's food delivery industry. Initially starting in Bangalore, Swiggy has expanded its operations to over 500 cities, revolutionizing the food delivery experience with its robust technology platform and reliable service.

Swiggy's growth trajectory has been remarkable. From humble beginnings, the company has raised substantial funding from prominent investors, reflecting strong market confidence. As of 2023, Swiggy's valuation stands at approximately \$10.7 billion, making it one of India's most valuable startups. This financial strength has enabled Swiggy to invest heavily in technology, logistics, and customer service, ensuring seamless and efficient food delivery.

Swiggy's mission is to provide convenient, reliable, and efficient food delivery services to customers while empowering local restaurants to reach a broader audience. With a network of dedicated delivery partners and a diverse range of restaurant partners, Swiggy ensures a wide variety of choices for consumers.

However, with growth come challenges, and it is essential to continuously analyze and address any operational issues to maintain customer satisfaction and loyalty."

3. Problem Statement:

3.1 Problem 1: Low Ratings in Specific Areas

- **Observation:** Certain areas have restaurants with consistently lower average ratings compared to the overall mean average rating.
- **Impact:** Low ratings in specific areas can affect the brand reputation and customer retention in those regions.

3.2 Problem 2: Low Ratings with Longer Delivery Times

- **Observation:** Restaurants with delivery times greater than 30 minutes receive predominantly low average ratings (below 3.0).
- **Impact:** Long delivery times negatively impact customer satisfaction and can lead to a decrease in repeat business and overall ratings.

3.3 Problem 3: Price Sensitivity and Customer Perception

- **Observation:** Investigating if there is a correlation between higher prices and lower average ratings.
- **Impact:** Understanding how pricing affects customer satisfaction can help in optimizing pricing strategies to improve value perception.

4. Background of Problems:

Problem 1: Low Ratings in Specific Areas

Low ratings in specific areas can be attributed to various factors such as poor service quality, inadequate restaurant facilities, or unfavourable local conditions. Identifying and addressing these factors is crucial for improving customer satisfaction in those areas.

Problem 2: Low Ratings with Longer Delivery Times

Timely delivery is a critical factor in the food delivery industry. Long delivery times can lead to cold or stale food, causing customer dissatisfaction. Analyzing the reasons behind long delivery times and finding ways to optimize the delivery process is essential for enhancing customer experience.

Problem 3: Price Sensitivity and Customer Perception

Customers' perception of value for money plays a significant role in their overall satisfaction. Higher prices without corresponding quality can lead to dissatisfaction. Understanding how price affects customer ratings can help in adjusting pricing strategies to better meet customer expectations.

5. Problem Solving Approach:

Details about the Methods Used with Justification

1. Data Exploration and Cleaning:

- **Method:** Initial exploration of the dataset involves understanding its structure, identifying missing or incorrect values, and handling outliers.
- **Justification:** Data cleaning is a critical first step to ensure the dataset is accurate and reliable. Removing outliers and handling missing values ensures that subsequent analyses are not skewed or misleading.

2. Descriptive Statistics and Visualization:

- **Method:** Calculating mean, median, and standard deviation for numerical variables and creating visualizations such as scatter plots, histograms, and bar charts.
- **Justification:** Descriptive statistics summarize the central tendency and variability of the data, providing a foundational understanding. Visualizations help in identifying patterns, trends, and potential anomalies, making it easier to interpret the data.

3. Correlation Analysis:

- **Method:** Computing the correlation matrix to identify relationships between variables such as price, average ratings, total ratings, and delivery time.
- **Justification:** Correlation analysis helps in understanding how variables interact with each other. Identifying strong correlations can highlight potential areas of concern or opportunities for improvement.

4. Segment Analysis:

- **Method:** Grouping the data by relevant categories (e.g., area, food type) to identify segments with specific issues and performing detailed analysis on each segment.
- **Justification:** Segment analysis allows for a more granular examination of the data, identifying unique patterns or problems within specific categories. This targeted approach helps in developing more effective and specific interventions.

Details about the Intended Data Collection with Justification

1. Source of Data:

- **Method:** The dataset was sourced from Kaggle, containing restaurant information such as ID, area, city, restaurant name, price, average ratings, total ratings, food type, address, and delivery time.
- **Justification:** Using a publicly available dataset from Kaggle ensures the data is comprehensive and relevant for the analysis. The dataset provides a rich source of information to identify and analyze various factors affecting customer satisfaction.

2. Data Collection Technique:

- **Method:** The dataset was created through web scraping by the original uploader on Kaggle.
- **Justification:** Web scraping is a common technique to gather large amounts of data from online sources. It allows for the collection of up-to-date and detailed information about restaurants, which is crucial for accurate analysis and insights.

Details about the Analysis Tools with Justification

1. Python and Pandas:

- **Method:** Utilizing Python programming language and the Pandas library for data manipulation and analysis.
- **Justification:** Python, along with Pandas, is widely used in data science due to its robust data handling capabilities. Pandas provides powerful tools for data cleaning, manipulation, and analysis, making it suitable for handling large datasets efficiently.

2. Visualization Libraries (Matplotlib and Seaborn):

- **Method:** Using Matplotlib and Seaborn libraries for creating visualizations.
- **Justification:** Matplotlib and Seaborn are popular Python libraries for data visualization. They provide a wide range of plotting functions that can be customized to create informative and visually appealing charts, helping in better data interpretation.

3. Statistical Analysis:

- **Method:** Applying statistical methods to analyze data distributions, correlations, and other relevant metrics.
- **Justification:** Statistical analysis is essential for understanding the underlying patterns and relationships in the data. It helps in making data-driven decisions and validating the findings through rigorous quantitative methods.

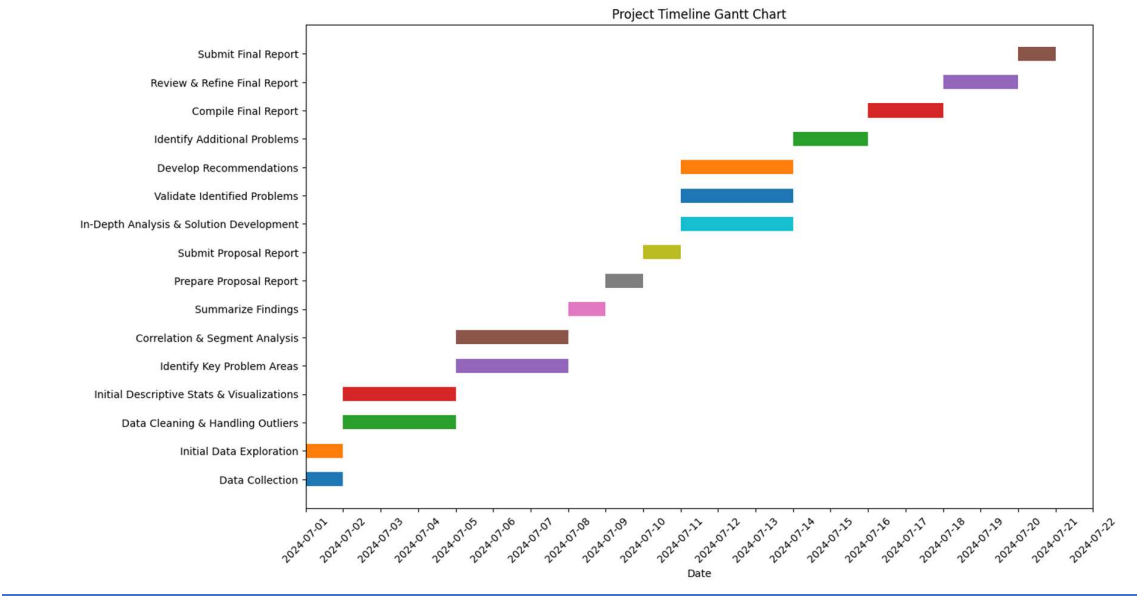
By employing these methods and tools, the analysis aims to provide a comprehensive understanding of the factors affecting customer satisfaction in Swiggy's operations and propose actionable solutions to enhance their service quality and overall customer experience.

6. Expected Timeline

Date	Activity
1st July	Data collection from Kaggle and initial data exploration.
2nd - 4 th July	Data cleaning, handling missing values, and removing outliers. Initial descriptive statistics and visualizations.
5th - 7th July	Identify key problem areas (e.g., low ratings in specific areas, impact of delivery time on ratings). Conduct correlation and segment analysis.
8th July	Summarize findings and identify the first problem related to low average ratings in specific areas.

9th July	Prepare the proposal report, detailing initial findings, problem statements, and proposed analysis methods.
10th July	Submit the proposal report.
11th - 13th	Conduct deeper analysis to validate identified problems. Develop actionable recommendations for each problem area.
14th - 15th July	Identify additional problems (e.g., relationship between delivery time and ratings, price sensitivity) and propose solutions.
16th - 17th July	Compile all findings, visualizations, and recommendations into the final report. Ensure clarity and coherence in the presentation of data and insights.
18th - 19th July	Review and refine the final report. Add executive summary, organizational background, and detailed methodology sections.
20th July	Final review and submission of the report.

Gant Chart:



7. Expected Outcome:

The project aims to deliver actionable insights and recommendations to Swiggy, focusing on enhancing operational efficiency and improving customer satisfaction. The following are the expected outcomes:

1. Identification of Key Problem Areas

- **Outcome:** A detailed identification of areas (specific regions, types of restaurants, etc.) with significantly lower average ratings than the overall mean.
- **Impact:** Helps Swiggy pinpoint regions or types of restaurants that need immediate attention, enabling targeted improvement strategies.

2. Insights into Delivery Time and Customer Satisfaction

- **Outcome:** A comprehensive analysis revealing the correlation between delivery times and customer ratings. Specifically, highlighting that restaurants with delivery times greater than 30 minutes predominantly receive low average ratings.
- **Impact:** Provides evidence for Swiggy to optimize delivery processes and reduce delivery times, thereby enhancing customer satisfaction and increasing repeat business.

3. Understanding Price Sensitivity

- **Outcome:** Insights into the relationship between pricing and customer ratings, potentially indicating that higher prices might be linked to lower average ratings.
- **Impact:** Guides Swiggy in adjusting pricing strategies to better align with customer expectations and perceived value, ensuring competitive pricing without compromising on quality.

4. Segment-Specific Analysis

- **Outcome:** Detailed segment analysis to understand unique issues faced by different categories, such as specific food types, city-specific challenges, and restaurant profiles.
- **Impact:** Enables Swiggy to implement tailored solutions for different segments, ensuring a more personalized and effective approach to improving service quality.

5. Actionable Recommendations

- **Outcome:** Clear, data-driven recommendations to address identified problems. This includes optimizing service quality in specific areas, improving delivery efficiency, and adjusting pricing strategies.
- **Impact:** Provides Swiggy with practical steps to enhance customer satisfaction, streamline operations, and maintain its competitive edge in the market.

6. Enhanced Data-Driven Decision Making

- **Outcome:** Establishment of a robust analytical framework that Swiggy can use for continuous monitoring and improvement.
- **Impact:** Fosters a culture of data-driven decision making within Swiggy, enabling ongoing improvements and quick adaptation to changing customer needs and market conditions.

7. Comprehensive Report

- **Outcome:** A well-documented report detailing the analysis process, findings, and recommendations.
- **Impact:** Serves as a valuable resource for Swiggy's management team, providing them with in-depth insights and a clear roadmap for operational improvements.

By achieving these outcomes, the project will help Swiggy improve its overall service quality, increase customer satisfaction, and strengthen its position in the highly competitive food delivery market.

